STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Course Outline

Chapter 6: Graphical tools to display data, Sections 6.1–6.4.

Chapter 1: Set theory, counting rules, Sections 1.1–1.3.

Chapter 2: Probability rules, Sections 2.1–2.4.

Chapter 3: Discrete random variables and their probability distributions, Sections 3.1–3.8.

Chapter 4: Bivariate distributions, Sections 4.1–4.5.

Chapter 5: Joint probability distributions, Sections 5.1–5.3.

Chapter 7: Estimation, Sections 7.1–7.4.

Chapter 8: Hypothesis testing, Sections 8.1–8.7.

Chapter 9: Regression + Correlation Analysis, Sections 11.1–?.

What Statistics Does

Statistics is the science of:

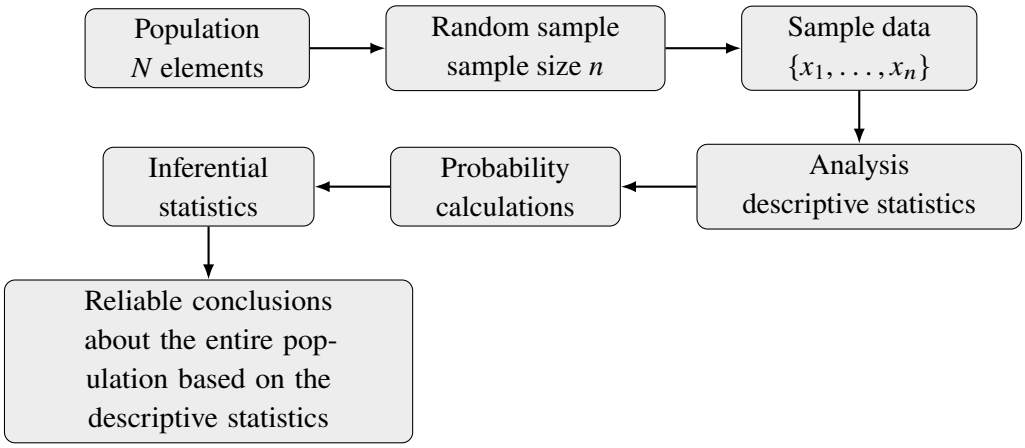
- collecting data/information;
- analyzing, graphing, summarizing, and organizing data/information — **descriptive statistics**;
- drawing reliable conclusions from descriptive statistics — **inferential statistics**.

1 Section 6.1 — A Typical Statistical Study

In a typical statistical study, we have a **population** of size N, from which a **random sample** of size n is drawn. The sample data are

$$\{x_1, x_2, \ldots, x_n\},$$

where x_i is the ith measurement/observation, for $i = 1, 2, \ldots, n$.



Homework: Exercise 6b (p. 213).
Lecture plan: Today — Sections 6.1–6.3; Wednesday — Section 6.4 (Boxplots).

2 Section 6.2 — Graphical Tools to Display Data

A **variable** is a characteristic that varies from one individual/element to another.

2.1 Examples and Classification

Examples:

1. a person's age (in years);
2. a student's GPA;
3. the country in which you are born.

Object	Classification	Description
Variable	numerical	Age; GPA
Variable	non-numerical	Country of birth
Dataset	numerical	Quantitative
Dataset	non-numerical	Qualitative / categorical

2.2 Graphing Non-Numerical Datasets

Example. In a class there are 40 students. The variable is the political-party affiliation of a student, with categories Democratic (D), Republican (R), and Other (O). A schematic dataset is

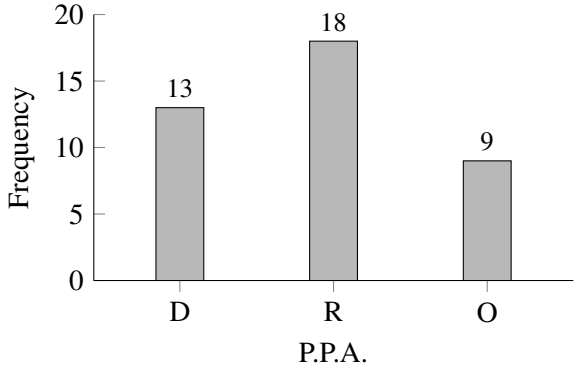
$$\{D, R, R, O, \ldots, R\}.$$

A standard workflow is

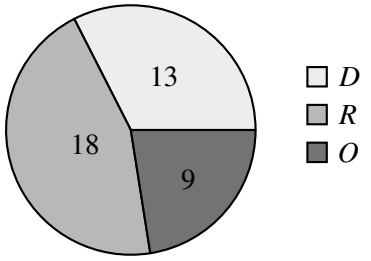
$$\text{Data} \longrightarrow \text{Table} \longrightarrow \text{Graph}.$$

Here, the table is a frequency distribution, displayed as a bar graph or pie chart.

P.P.A.	Frequency	Relative frequency	Pie slice
D	13	13/40	$\frac{13}{40}(360°) = 117°$
R	18	18/40	$\frac{18}{40}(360°) = 162°$
O	9	9/40	$\frac{9}{40}(360°) = 81°$
Total	40	1	360°



(a) Bar graph using the recorded frequencies.

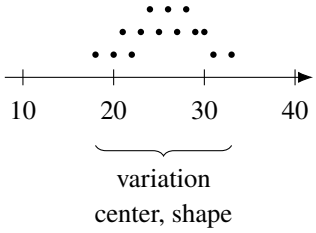


(b) Pie chart using the same 13 : 18 : 9 data.

2.3 Graphing Numerical Datasets

1. Dotplot (small datasets)

A dotplot is useful for seeing **variation**, **center**, and **shape**. The schematic cluster below illustrates these features on a scale from roughly 10 to 40.



2. Stem-and-Leaf Plot (small datasets)

Notes on Exercise 8(a): the **stem is hundredths** and the **leaf is thousandths**.

Stem	**Leaf**
02	0
03	0 0
04	4 4 6 9 0
05	2 2
06	6
07	

A more developed example is

Stem	Leaf
02	0
03	0 0 7 9 9 9
04	0 2 2 4 4 5 6 7 9
05	0 1 1 2 2 3 4 5 6 7 8
06	0 1 1 1 2 2 6
07	

A stem-and-leaf plot must specify the place values of its stem and leaf.

Stem	Leaf
1	0 0 6 9
2	1 7
3	4 4 5 8 8

Without a place-value key, an entry such as 3 | 5 is ambiguous: it could represent 0.035, 0.35, 3.5, or 35.

3. Histogram

Homework 8(d) — Histogram: use $[0.02, 0.03)$ as the first interval.

About **5–10 intervals** are typically used. The workflow is again

$$\text{Data} \longrightarrow \text{Table} \longrightarrow \text{Graph},$$

A frequency distribution then leads to the histogram.

Level	Frequency	Relative frequency
$[0.02, 0.03)$	1	$1/35$
$[0.03, 0.04)$	6	$6/35$
$[0.04, 0.05)$	9	$9/35$
$[0.05, 0.06)$	11	$11/35$
$[0.06, 0.07)$	7	$7/35$
$[0.07, 0.08)$	1	$1/35$
Total	$n = 35$	1

Rounded relative frequencies may sum to a value close to 1 (for example, 0.99 or 1.01).

An alternative choice of classes is

$$[0.020, 0.028)\,, \qquad [0.028, 0.036)\,, \ldots$$

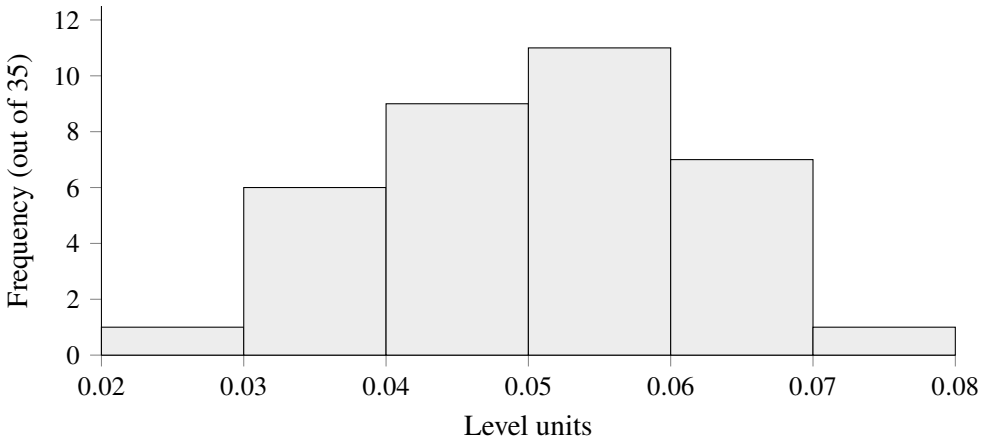
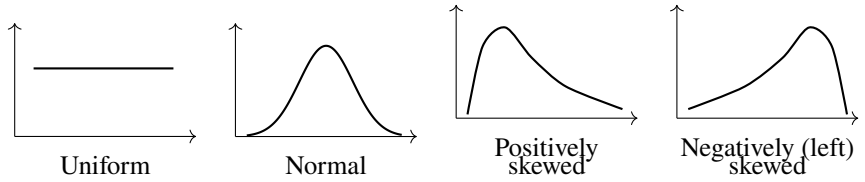


Figure 2: Frequency histogram from the interval table.

Distribution Shapes

Common distribution shapes include the following.



Ogive (R.C.F.)

An ogive displays relative cumulative frequency.

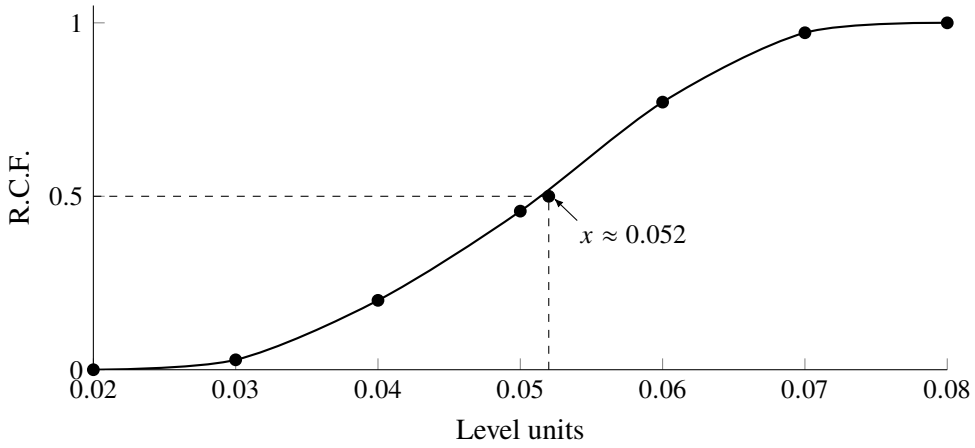


Figure 3: Ogive constructed from the frequency table.

Homework: 10(a,d,e), using $[5.0, 6.0)$ as the first interval; 16, using $[0.0, 0.4)$.

3 Section 6.3 — Numerical Measures for Numerical Datasets

3.1 I. Measures for the Center

1. Sample Mean/Average

Notation: $\bar{x}$.

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}.$$

The summation symbol $\sum$ is capital sigma.

For the data $\{1, 6, 10, 9\}$,

$$\bar{x} = \frac{1 + 6 + 10 + 9}{4} = \frac{26}{4} = 6.5.$$

The population mean/average is denoted by μ:

$$\mu = \frac{x_1 + x_2 + \cdots + x_N}{N}.$$

2. Sample Median

Notation: $\widetilde{x}$.

- First order the dataset.
- If n is odd, $\widetilde{x}$ is the middle observation in the ordered dataset.
- If n is even, $\widetilde{x}$ is the average of the two middle observations.

  Example:

$$\begin{array}{ccccc} 1 & 5 & 6 & 9 & 10 \\ 1 & 2 & 3 & 4 & 5 \end{array} \qquad \Longrightarrow \qquad \widetilde{x} = 6.$$

Resistance to Outliers

The sample median is an example of a **resistant measure**: it is not strongly influenced by extreme outliers in a dataset. The sample mean is **not resistant**.

$$\{1, 2, 3, 4, 5\}: \qquad \bar{x} = \widetilde{x} = 3,$$

whereas

$$\{1, 2, 3, 4, 50\}: \qquad \bar{x} = 12, \qquad \widetilde{x} = 3.$$

Next Lecture — Section 6.4: Boxplots

We will cover Section 6.4, **Boxplots**, in the next lecture on Wednesday.